Teaching Tech Together

Finale

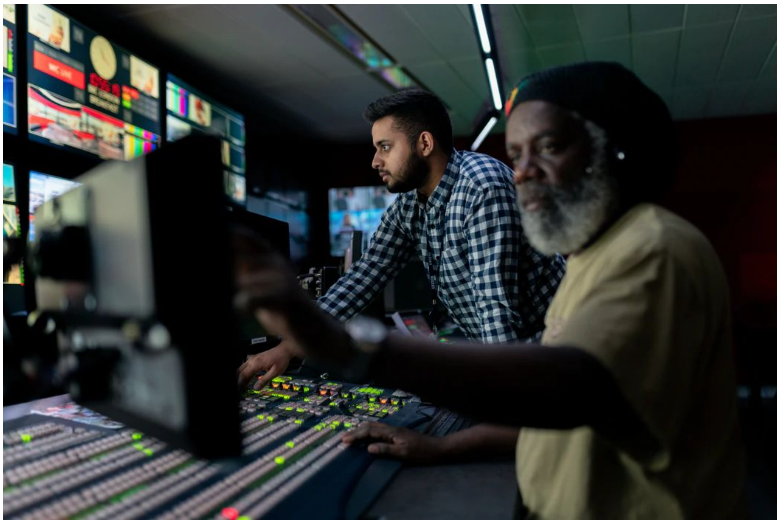
Greg Wilson
https://third-bit.com

Estimated time: 40 minutes.
All art from openclipart.org (open license)

Learning Objectives

1. Explain the difference between active and passive learning and give two explanations for why the former is more effective.
2. List six strategies learners can use to accelerate their learning and justify at least two of them in terms of the ideas introduced in this training
3. Identify several common myths about education and learning.

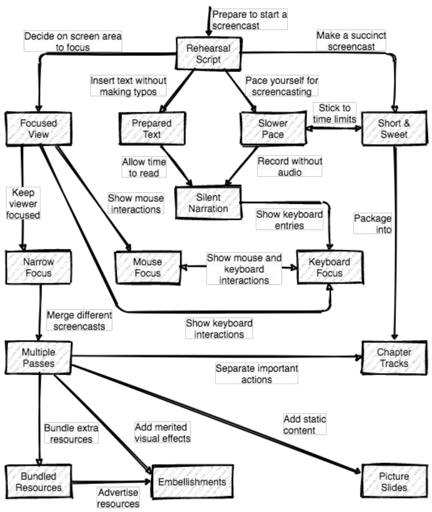
Don’t try to be (or beat) recorded video



Requires time and money

Don't try to be (or to beat) recorded video. Even mediocre videos require time and production experience that you probably don't have right now, so move your live teaching online instead of trying to record and edit yourself.
https://unsplash.com/photos/t4qI2IDcL5s

But if you do…



Chen & Rabb:
“A Pattern Language for Screencasting.”
https://doi.org/10.1145/1943226.1943234

If you are going to make videos, please read Chen and Rabb's "A Pattern Language for Screencasting" first.
https://doi.org/10.1145/1943226.1943234

Don’t teach to empty chairs

This is hard

Don't teach to empty chairs. We all rely on small social cues like nods and "mm hms" far more than we realize. Teaching without these is really hard, so if it's not practical to have everyone turn on video, choose a subset of 4-6 learners and have them do it so that your hindbrain believes you actually have an audience. This also promotes engagement: learners can see that they're not alone, and after each break, you can select a different subset of learners and direct your questions to them.
https://unsplash.com/photos/3XSyPucMF3c

Add speaker’s notes to your slides

School is difficult enough

Add speaker's notes to your slides. These can be point-form or long-form depending on how much time you have, but they will make your content easier to find, easier to review, and more accessible to people who have trouble seeing or hearing.

Invest in a good microphone

Get it away from your face and keyboard

Your voice matters more than your face. You can switch off video if you have to, but if you lose sound, your class is over. You can use a headset with a microphone, but this can become uncomfortable after a couple of hours (particularly if you wear glasses). A Yeti or Samson Meteor microphone is more comfortable, but if you use one, it's worth $30 to get a stand (because if it's on your desk, all your audience will hear is your typing).

Share work in real time

HackMD or Google Doc

https://unsplash.com/photos/Wh9ZavMYeIw

Share work in real time. Have learners share their work in a Google Doc or HackMD during the lesson. (This is no more or no less intimidating than asking a music student to play something for the class.) When you do this, paste in a list of people's names so that they know where to type. Once they're comfortable with this (which will take a few hours), have them take shared notes in the same doc so that your copilot can see what they think you're saying.

Keep track

- Yellow: names

https://unsplash.com/photos/gdL-UZfnD3I

Keep track of who has and hasn't spoken. Again, a laissez faire class can quickly turn into a discussion among a couple of extroverts, so keep a tally of who's spoken how often and try to even it out. You don't need technology to do this: a scrap of paper and a pencil will work just fine.

Keep track

- Yellow: names
- Red/green: status

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Keep track

- Yellow: names
- Red/green: status
- Blue: breaks

https://unsplash.com/photos/gdL-UZfnD3I
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Two Ways to Learn

Read about something	Try it out
Watch a video	Do exercises
Attend a lecture	Discuss a topic
Hear something explained	Try to explain it

We’ve been talking about how teachers can build better lessons. We’re now going to shift our focus to how to deliver those lessons. Our starting point is the contrast between two styles of lesson delivery. Take a moment and read these two lists, then vote for the style that you think works best.

Two Ways to Learn

Read about something	Try it out
Watch a video	Do exercises
Attend a lecture	Discuss a topic
Hear something explained	Try to explain it
Passive Learning	**Active Learning**

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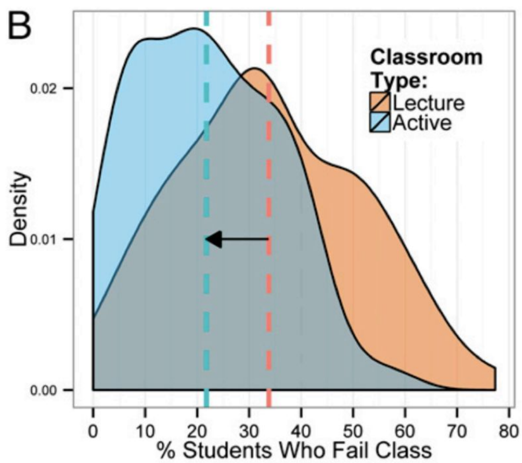
It’s probably no surprise that *active learning* outperforms *passive learning*.

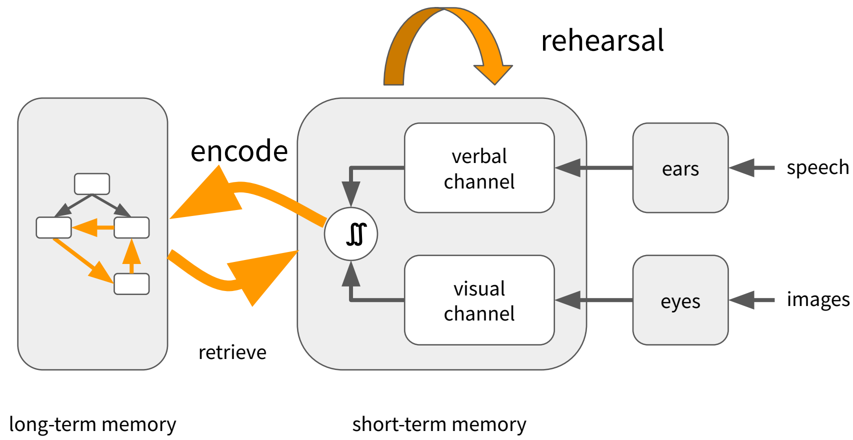
Active Learning is Better



…but university students report that they prefer passive learning…

For example, this plot shows reduction in failure rates in STEM classes (from http://www.pnas.org/content/111/23/8410.full.pdf), and we now have the cognitive science to explain why. However, learners *report* that they prefer passive learning, even though active learning works better (https://www.pnas.org/content/116/39/19251)

Why?



The longer something stays in short-term memory, the greater the chance that it will be encoded into long-term memory. Having people *rehearse* information therefore helps with retention. Similarly, the more practice people have retrieving information, the more links will form in their mental model, and the stronger those links will be. Having them recall information and use it therefore also increases learning.

Six Strategies

http://www.learningscientists.org/

There are six strategies that you and your learners can use to leverage all of this, and the best summary of them comes from the Learning Scientists website. We'll go through them one by one.

Spaced Practice

- 5x2 hours is better than 2x5 hours
- Include some older material in every new lesson
- One of the few advantages of traditional lecture-based classes over just-in-time online courses

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The first strategy is *spaced practice*. Five two-hour study sessions are more effective than two five-hour sessions, and far more effective than one 10-hour cramming session. While you can't control your learners' study habits, you can include some older material in every new lesson. (Spacing things out may be one of the few advantages of traditional lecture formats over just-in-time online learning.)

Retrieval Practice

- Practice using information…
- …in the context in which it will actually be used
- Repeated testing improves recall
- Program in order to learn how to program

The second strategy is *retrieval practice*. This one seems obvious—you'll be better at remembering things if you practice remembering them—but it's important to practice remembering in a realistic context. If you want to recall information for a multiple-choice quiz, practice doing multiple-choice quizzes; if you want to remember syntax rules when programming, practice recalling them as you're programming.

Elaboration

- Explain things in detail to yourself
- Explain them to others

The third strategy is *elaboration*. We all know that teaching something is a great way to learn it, and in general, explaining things to yourself or to someone else is a good way to strengthen your understanding. One way to implement this is to follow up every question on a practice quiz with a detailed (self-)explanation of why that’s the right answer. Another is to compare and contrast new information with old.

Interleaving

- A-B-C-A-B-C is better than A-A-B-B-C-C
- A-C-B-C-A-B is better still

The fourth strategy is *interleaving*. Mixing up the study of different topics improves later recall because it builds more long-range links in your mental model; randomizing the order is better than a pattern. (Think about the lyrics to a song: if you always practice in one order, you'll only be able to recall in that order.)

Concrete Examples

- Novices don’t know enough yet to make generalizations concrete
- So provide examples…
- …and be specific about the general principles each example embodies

The fifth strategy is *concrete examples*. Novices (and even competent practitioners) may not know enough to be able to apply a general principle to a specific case, so give them examples of that. Similarly, whenever you solve a specific problem, take a moment to describe the general principles used.

Dual Coding

- Combine words and visuals

The final strategy is *dual coding*, which we discussed earlier. Images and words are more effective in combination than either on its own.

Exercise

Give yourself a score of 0-6 on your use of these learning strategies.

Spaced practice	Retrieval practice
Elaboration	Interleaving
Concrete examples	Dual coding

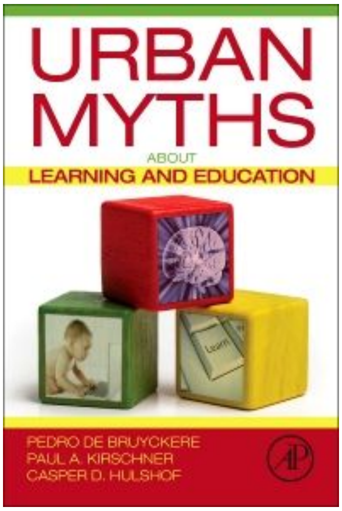
Which of the learning strategies do you use?

Helping disparate learners

1. Avoid if you can
2. Split the room
3. Use advanced learners as helpers
4. Use pair programming (for everyone)
5. Synchronous self-paced work

Here are five strategies for handling cases where you have learners of widely different levels in the same class.

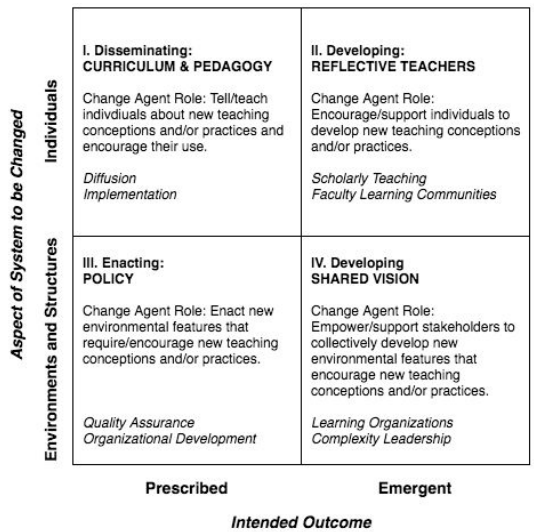
Learning Styles are Not a Thing



- Neither is stereotype threat
- Or fixed vs. growth mindset
- And something else in this course is probably wrong as well: we just don't know what (yet)

"Learning styles" are definitely not a thing. Stereotype threat and fixed versus growth mindset probably aren't, but the science is still evolving.

Institutional Change



From Borrego & Henderson, "Increasing the Use of Evidence-Based Teaching in STEM Higher Education", 2014

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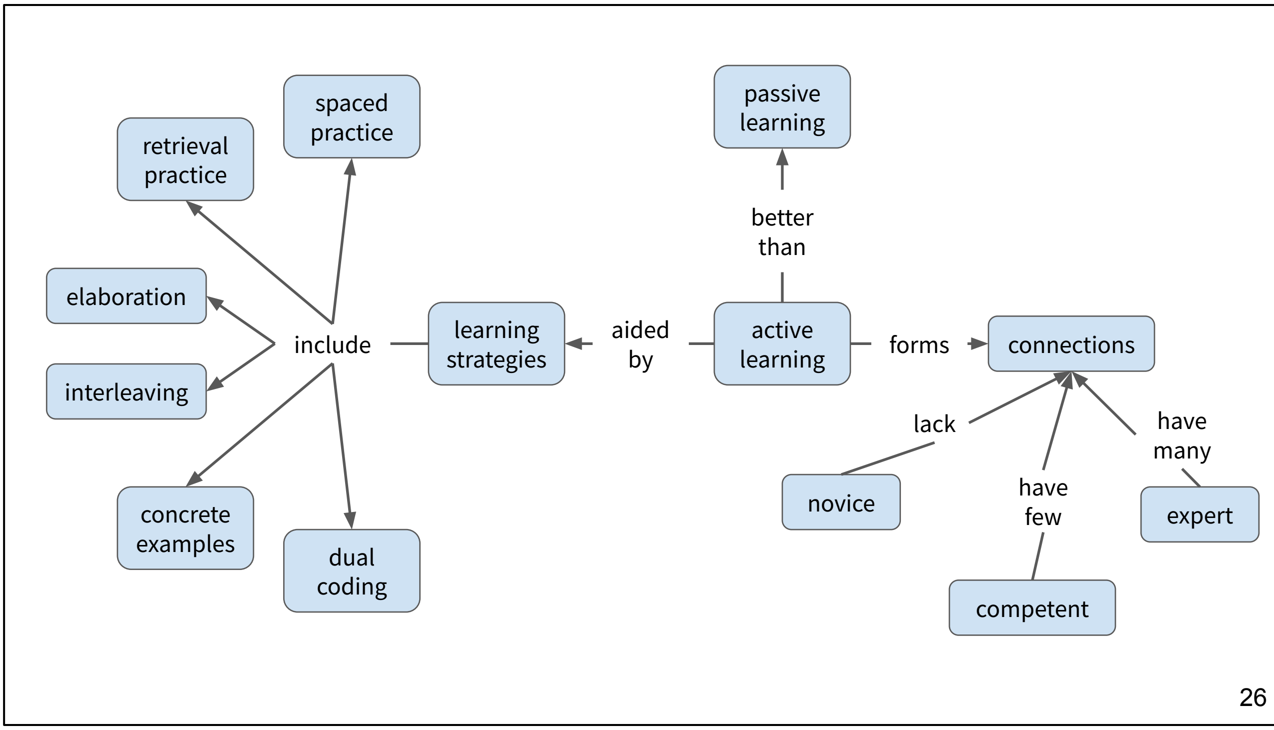

Concept map for learning strategies.

What Next?

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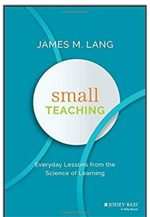
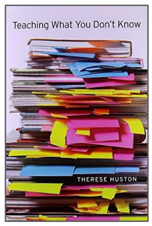
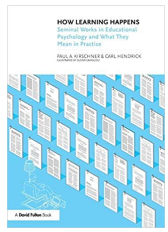
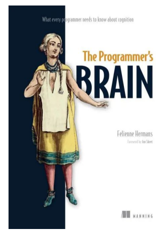
Critics sometimes say that online education is what everybody wants for somebody else's children. The data backs that up: while they can be effective for some learners in some situations, MOOCs still have very low completion rates, are often pedagogically ineffective, and can increase learners' feeling of isolation. But that doesn't mean that all online education is a bad idea. Synchronous online collaboration is now normal for children, gamers, and office workers. It's easy to imagine a world in which video conferencing tools support peer instruction, and in which sharing your programming session with the rest of the class is as routine as sharing a picture of your cat.

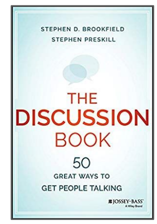
Where Next?

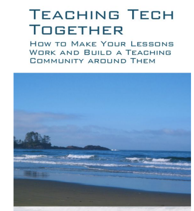












Lang: *Small Teaching*

Small Teaching looks at what you can do if you don't have the time or budget to do the absolutely best thing.
https://www.amazon.com/Small-Teaching-Everyday-Lessons-Learning/dp/1118944496/

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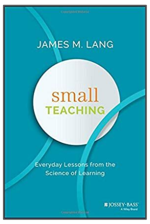

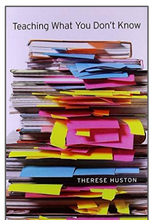

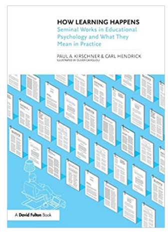

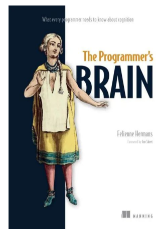

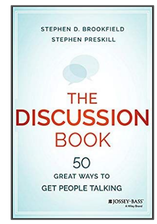

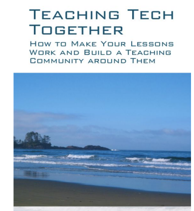

Huston: *Teaching What You Don't Know*

Teaching What You Don't Know explains how theory can help you in practice when you're only two pages ahead of your learners.
https://www.amazon.com/Teaching-What-You-Dont-Know/dp/0674066170/

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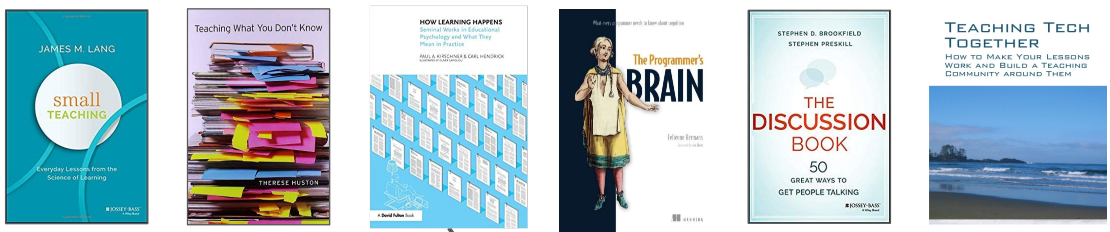

Kirschner & Hendrick: *How Learning Happens*

How Learning Happens summarizes key education research papers and explains why they matter.
https://www.amazon.com/How-Learning-Happens-Paul-Kirschner/dp/0367184575/

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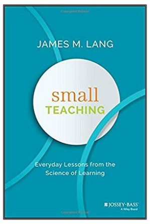

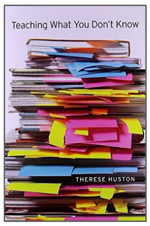

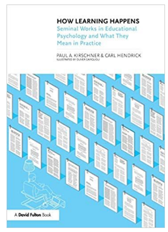

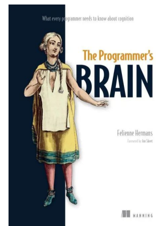

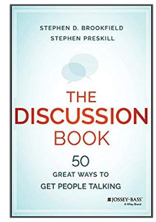

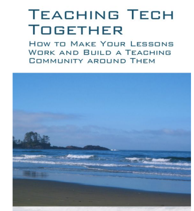

Hermans: *The Programmer's Brain*

Urban Myths catalogs a lot of things that people believe that *aren't* true.
https://www.amazon.com/Urban-Myths-about-Learning-Education/dp/0128015373/

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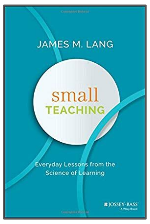

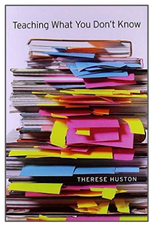

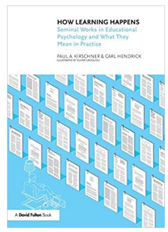

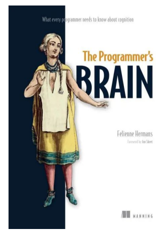

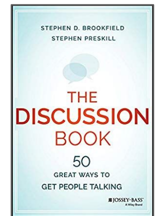

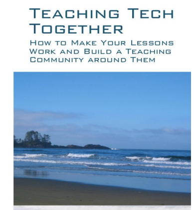

Brookfield and Preskill: *The Discussion Book*

The Discussion Book catalogs 50 ways that people can share ideas and make decisions.
https://www.amazon.com/Discussion-Book-Great-People-Talking/dp/1119049717/

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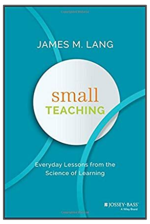
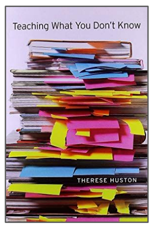
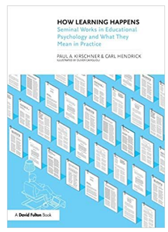
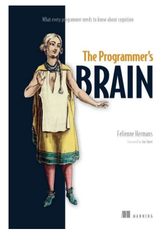
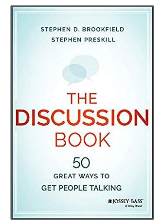
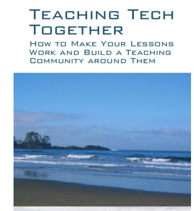








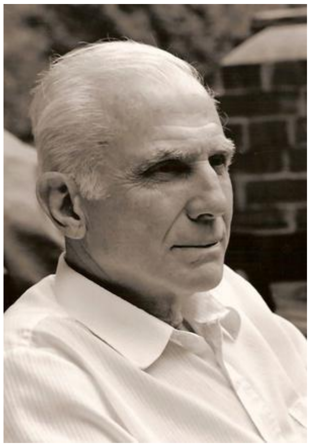




http://teachtogether.tech

Finally, *Teaching Tech Together* is a work in progress that tries to bring key elements of theory and practice together for people with technical backgrounds who find themselves teaching in free-range settings. All of its content is freely available online. http://teachtogether.tech/

Thank You



Start where you are.

Use what you have.

Help who you can.

These three rules aren't the whole of teaching, but they're a good start. Thank you for listening.